Ransomware Early Warning

Android Ransomware-Like Activity Early Warning in a Controlled Lab

Detect suspicious file-system behavior, measure the warning, and stop only the controlled test workload.

ENVIRONMENT

1 emulator/device

SCOPE

1 test directory

EVALUATION

2 benign + 3 simulated

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PROBLEM FRAMING

Unknown ransomware can change files before signatures catch up

SIGNATURE GAP

New samples can evade known signatures

Signature-based antivirus may miss zero-day or uncatalogued ransomware. Recognition can arrive after file changes have begun.

ANDROID BOUNDARY

Apps cannot inspect everything

Android sandboxing prevents ordinary cross-app memory inspection. A normal app cannot promise device-wide visibility or control.

OBSERVABLE CLUE

Destructive activity leaves a trail

Bursts of open, write and rename operations across many files provide a behavioral signal worth testing.

Monitor the designated directory, raise an early warning, and terminate only the registered test workload.

Dummy files only. No real malware. No device-wide prevention claim on stock Android.

BEHAVIORAL RATIONALE

File changes expose the behavior we need to measure

Ransomware commonly makes files unavailable through encryption or other changes. The prototype asks whether a workload is changing many files unusually quickly in a suspicious sequence.

01

Open

Observe access to files inside the explicitly accessible test directory.

02

Write

Measure modification bursts and the number of distinct files touched.

03

Rename

Track rename activity alongside write rates and event ordering.

FileObserver / inotify is the telemetry candidate. Verify event coverage in the chosen emulator or rooted device before evaluating the detector.

The event pattern is evidence for a heuristic, not proof of malicious intent. Benign bulk updates can produce similar activity.

Platform documentation defines both the method and its limits

REFERENCE	WHAT IT ESTABLISHES	DESIGN CONSEQUENCE
Android FileObserver API	Documents inotify-backed file and directory events. It does not guarantee device-wide monitoring.	Collect scoped events and verify coverage.
MITRE ATT&CK T1486	Data Encrypted for Impact describes behavior that renders files inaccessible.	Model file-impact behavior with harmless simulations.
Android security best practices Android app sandbox	Permissions and application isolation constrain cross-app inspection and control.	Respect sandbox boundaries and document lab privileges.

The contribution is a measurable, explainable lab pipeline

Evaluate warning quality and file impact using benign workloads and safe simulations, without treating threat taxonomy as evidence of prototype performance.

DETECTOR DESIGN

Start with transparent rules that can be audited and tuned

A rolling feature window summarizes activity in the designated directory. Combine signals rather than treating every bulk update as ransomware-like behavior.

ACTIVITY RATE

Write and rename rates

Count observed operations over the window to identify unusually concentrated file changes.

BREADTH

Distinct files touched

Distinguish repeated changes to one file from changes spread across many files.

PATTERN

Event sequence

Evaluate open, write and rename ordering alongside rates and file counts.

DECISION

Explainable heuristic score

Apply transparent rules or weights, then calibrate thresholds against the two benign workloads.

On a trigger, log the alert and stop only the registered test workload. Machine learning remains an optional extension after a reliable baseline.

EXPERIMENTAL MATRIX · PROPOSED EVALUATION

Compare baseline and calibrated thresholds on the same five workloads

Run exactly 2 benign workloads and 3 harmless ransomware-like simulations repeatedly on one documented lab setup.

WORKLOAD COHORT	BASELINE CONFIGURATION	CALIBRATED CONFIGURATION
2 benign workloads	Single-event frequency threshold over a fixed time window; log alerts and misses	Combined rate, distinct file count and rename-burst heuristic; tunable rolling window
3 harmless ransomware simulations	Same threshold; record triggers and misses	Scoped, prompt workload termination on calibrated alert

Detection delay

Elapsed time from run start to alert trigger, in ms or s

Files changed before alert

Distinct impacted files prior to containment

Precision

True positive alerts divided by all triggered alerts

Recall

Detected simulation runs divided by all simulated attacks

Retain run labels and TP / FP / FN outcomes. Define the delay start point consistently; report missed runs separately. No thresholds or performance results are assumed in advance.

SYSTEM ARCHITECTURE · CONTROLLED AND AUDITABLE

The full response path stays inside the lab boundary

01

Execute workload

Launch one benign or harmless simulated run through the test harness.

02

Capture events

FileObserver / inotify watches only the designated target directory.

03

Window and score

Aggregate features and apply heuristic rate scoring.

04

Respond and evaluate

Log the alert, terminate only the registered test workload, and calculate metrics.

TELEMETRY BOUNDARY

One explicit lab directory

RESPONSE BOUNDARY

Harness-controlled workload only

Event logs, run metadata and ground-truth labels make each decision auditable. Directory events alone do not establish process ownership.

DATA PIPELINE · FROM RAW EVENTS TO EVIDENCE

Every alert must trace back to a labeled run

01 **Workload execution**

Record the benign or simulated label, run start time and registered workload identity.

02 **FileObserver / inotify capture**

Timestamp permitted file events from the designated target directory.

03 **Feature windowing and heuristic rate scoring**

Compute write and rename rates, distinct-file counts and event patterns; retain the configuration.

04 **Targeted termination and metrics evaluation**

Log the alert, stop only the registered simulation if triggered, and compare outcomes with ground truth.

Export JSON/CSV event logs, run summaries, reproducible metrics and demo evidence.

FOUR-PHASE DELIVERY

Advance through evidence gates, not assumed readiness

PHASE	IMPLEMENTATION WORK	EXIT GATE
Phase 1 Lab & Telemetry	Set up one emulator or rooted device, create the designated directory, and document event visibility.	Repeatable event logs
Phase 2 Detector Baseline	Implement rolling windows, transparent rules, alert logging and the scoped stop action.	Alert on crafted test events
Phase 3 Test & Tune	Run 2 benign workloads and 3 safe simulations; calibrate thresholds and repeat trials.	Four core metrics computed from logs
Phase 4 Package & Demo	Finalize diagrams, report, pseudocode, private repository and mentor walkthrough.	Reproducible end-to-end demo

Use short iterations. Do not begin detector evaluation until the telemetry source is verified.

Demonstrate the warning, the scoped stop, and the evidence

01

Establish the boundary

Show the chosen lab environment, designated directory and registered test workload.

02

Run the comparison

Walk through benign activity and a harmless simulation, then inspect the alert and targeted stop action.

03

Inspect the record

Review event logs, run labels, detection delay, files changed before alert, precision and recall.

Questions for the walkthrough

- Which events are visible in this setup?
- What causes false alerts or missed simulations?
- How is termination restricted to the test workload?

OFFICIAL REFERENCES

1. Android Developers: FileObserver
2. Android security best practices
3. MITRE ATT&CK: T1486
4. Android application sandbox

Proposed prototype and evaluation plan. Performance results will be added only after experiments; no device-wide protection claim is made.